

Transfer Manifest - Technology #14

Acoustic Obstruction Detection for CSF Shunts

YOU RECEIVE:

- + Acoustic obstruction detection concept + mechanism description
- + Acoustic impedance + reflection + TOA governing equations
- + Bench test protocols (detection sensitivity, false positives, acoustic output, EMC)
- + External precedent catalog (IVUS, Doppler, AIUM/NEMA UD-2)
- + Critical UNKNOWN disclosure (tissue obstruction sensitivity, optimal frequency, false positive rate)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production transducer design (frequency, geometry, material)
- > Production drive + receive electronics + signal processor
- > Production classifier (validated in vivo)
- > Manufacturing process for catheter-integrated acoustic detector
- > Regulatory submission (510(k) with IVUS predicate, or De Novo)
- > Clinical validation (sensitivity + specificity vs imaging)

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

NOT AVAILABLE:

- Physical prototype
- Clinical data
- Qualified supplier
- Granted IP
- Independent engineer evaluation

This manifest is honest. Items marked 'not available' do not exist.